Electric lighting

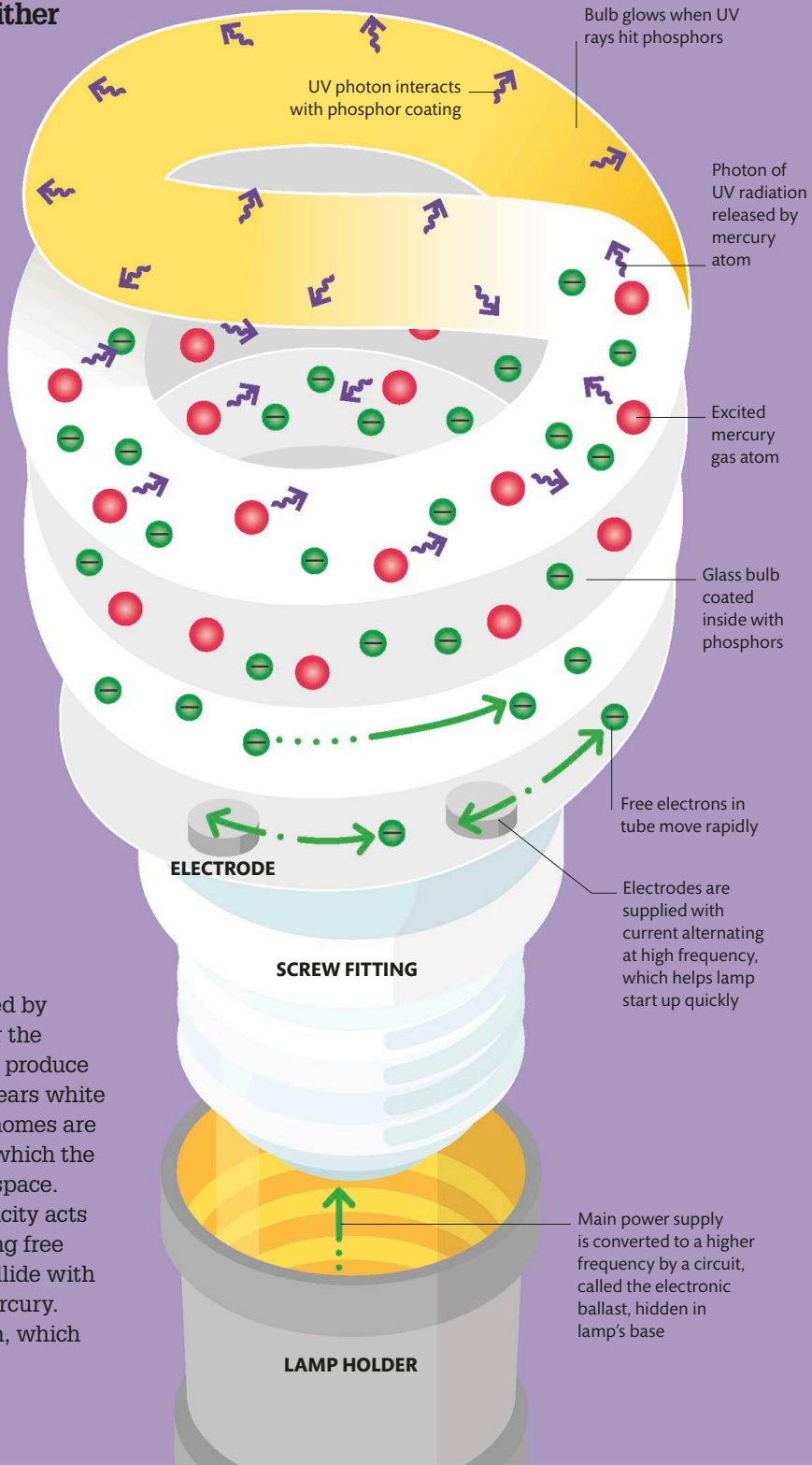
Most electric lighting uses either fluorescent or LED lamps (bulbs). Far less energy-efficient incandescent light bulbs can still be found, although their use is declining.

3 Visible light produced
When the ultraviolet radiation hits the phosphors painted on the glass, it causes them to glow. There are red, green, and blue phosphors, so the overall combination appears white.

2 Electrons release energy
The excited electrons "fall" back down to their original energy level. As they do so, they release energy in the form of ultraviolet radiation. This radiation is not visible to human eyes.

1 Electrons are excited
High-voltage electricity passes through low-pressure mercury vapor within the bulb. Electrons in the mercury atoms are excited, or knocked to a higher energy level.

KEY
- Free electron
- Excited mercury atom



Compact fluorescent lamps

In a fluorescent lamp, light is produced by pigments called phosphors that cover the inside of a glass tube. The phosphors produce red, green, and blue light, which appears white when combined. The lamps used in homes are compact fluorescent lamps (CFL), in which the tube is twisted around itself to save space. When the light is switched on, electricity acts on the vapor in the glass tube, exciting free electrons in the vapor so that they collide with other electrons bound to atoms of mercury. This creates ultraviolet (UV) radiation, which hits the phosphors, producing light.